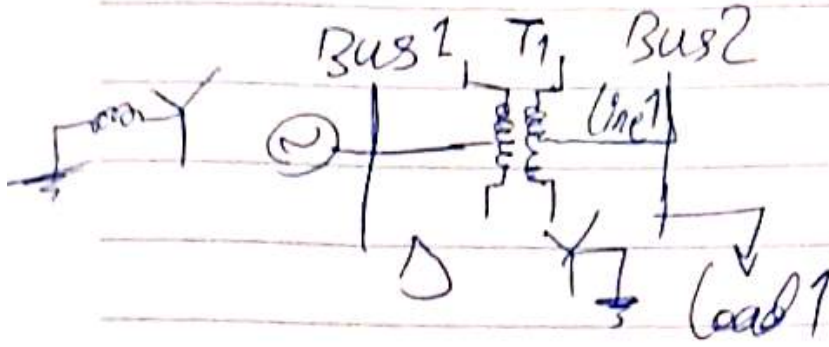


ارمین بنبرگری ۱۳۵۲/۱۰/۱۵
گزینه دوم قلیل

Date:

Sat Sun Mon Tue Wed Fri

Subject:



بخش الف

$$S_{base} = 50 \text{ MVA}$$

$$V_{base, Bus 1} = 9.4 \text{ kV} \Rightarrow V_{base, Load 1} = V_{base, Bus 1} \times \frac{9.4 \text{ kV}}{9.4 \text{ kV}}$$

$$= 9.4 \text{ kV} \times \frac{9.4 \text{ kV}}{9.4 \text{ kV}} = 9.4 \text{ kV}$$

$$X_{Pu, new} = X_{Pu, old} \times \left(\frac{V_{base, old}}{V_{base, new}} \right)^2 \times \left(\frac{S_{base, new}}{S_{base, old}} \right)$$

$$\Rightarrow X_{G1, Pu} = 0.1 \times \left(\frac{9.4}{9.4} \right)^2 \times \left(\frac{50}{50} \right)^2 = 0.1 \text{ pu}$$

$$X_{T1, Pu} = 0.1 \times \left(\frac{9.4}{9.4} \right)^2 \times \frac{50}{50} = 0.1 \times 0.911 \times 1.44 \text{ pu} = 0.13 \text{ pu}$$

$$Z_{base, 2} = \frac{(V_{base, 2})^2}{S_{base, new}} = \frac{(9.4 \text{ kV})^2}{50 \text{ MVA}} = \frac{88.36 \text{ kV}^2}{50} = 1.7672 \text{ pu}$$

$$\Rightarrow X_{Line, Pu} = \frac{X_{actual}}{Z_{base, 2}} = \frac{1}{1.7672} = 0.566 \text{ pu}$$

ص

Date:

1 2 3 4 5 6 7 8 9 10 11 12

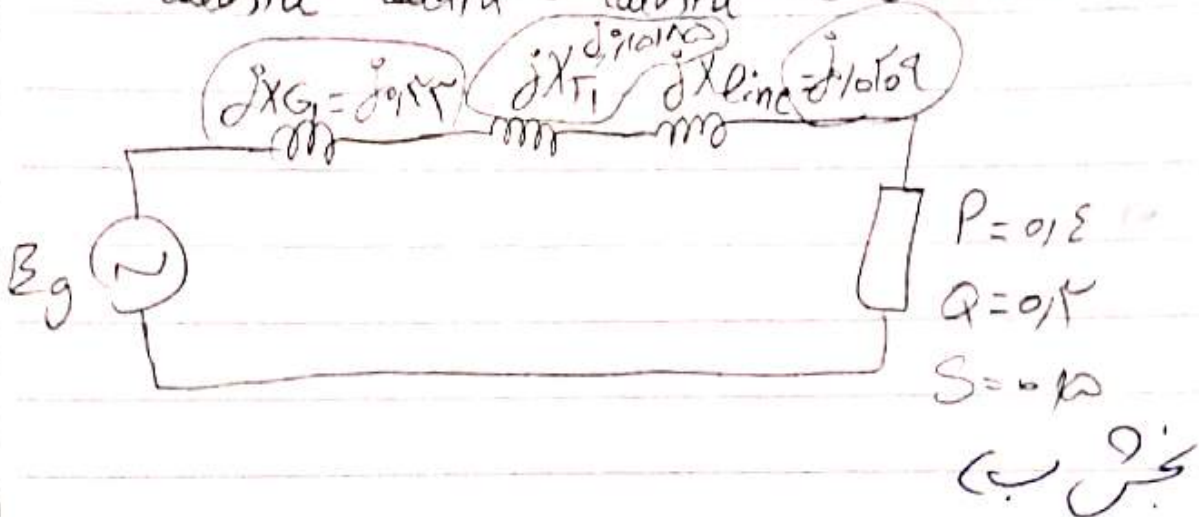
Subject:

Load $\Rightarrow 20 \text{ MVA}$, 0.9 lagging

$$P_{\text{load}, Pu} = \frac{P_0 \text{ MW}}{S_0 \text{ MVA}} = 0.9 Pu \Rightarrow \text{P.F.} = 0.9 \Rightarrow \tan(\phi) = \frac{0.4}{0.9} = 0.44$$

$$\Rightarrow Q_{\text{load}, Pu} = P_{\text{load}, Pu} \times \tan(\phi) = 0.9 \times 0.44 = 0.4 Pu$$

$$\Rightarrow S_{\text{load}, Pu} = P_{\text{load}, Pu} + jQ_{\text{load}, Pu} = 0.9 + j0.4 Pu$$



$$V_{\text{Bus2}} = V_r = 0.9 \text{ pu} \angle 0^\circ \text{ pu}, S_{\text{load}} = P + jQ = 0.9 + j0.4 Pu$$

$$Z_K = j(X_{T1} + X_{\text{line}}) = j0.10 + j0.05 = j0.15 \text{ pu}$$

$$S = V I^* \Rightarrow \frac{S}{V} = I^* \Rightarrow I = \left(\frac{S_{\text{load}}}{V_r} \right)^* = \frac{P - jQ}{V_r^*} = \frac{0.9 - j0.4}{0.9 \text{ pu}} = 1.0 - j0.44 \text{ pu}$$

Date: / /

Sat	Sun	Mon	Tue	Thu	Wed	Fri
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Subject: _____

$$V_1 = V_r + (\bar{I} \times Z_{1r}) = 0.1 \angle 0^\circ + (0.18 \angle 45^\circ - j0.15 \angle 90^\circ) \times j0.1 \angle 90^\circ$$

$$= 0.1 \angle 0^\circ + \underbrace{j(0.18 \angle 45^\circ \times 0.1 \angle 90^\circ)}_{-1} - j(0.15 \angle 90^\circ \times 0.1 \angle 90^\circ) = 0.1 \angle 0^\circ + j0.018 \angle 135^\circ$$

$$0.1 \angle 25.8^\circ = 1.0 \angle 45^\circ + j0.1 \angle 90^\circ P_u \Rightarrow |V_1| = \sqrt{(1.0 \angle 45^\circ)^2 + (0.1 \angle 90^\circ)^2}$$

$$= \sqrt{1.0^2 + 0.01} \angle 0.5^\circ = \sqrt{1.01} \angle 0.5^\circ \approx 1.005 \angle 0.5^\circ P_u \Rightarrow \theta_1 = \tan^{-1}\left(\frac{0.1 \angle 90^\circ}{1.0 \angle 45^\circ}\right)$$

$$\Rightarrow V_1 = 1.005 \angle 0.5^\circ P_u$$

$$25.8^\circ$$